

SAINT ANDREW'S JUNIOR COLLEGE

2008 PRELIMINARY EXAMINATION

MATHEMATICS

Higher 2

9740/01

Paper 1

Wednesday

27 August 2008

3 hours

Additional materials : Answer paper
List of Formulae(MF15)
Cover Sheet

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Answer **all** the questions. Total marks : 100

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically state otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematic steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This document consists of 5 printed pages including this page.

[Turn over

- 1 Show that $x^2 - 6x + 16 > 0$ for all $x \in \mathbb{R}$ and hence, solve the inequality

$$\frac{1}{6-x} \leq \frac{16}{x^3 - 12x^2 + 36x}. \quad [6]$$

- 2 Find the binomial expansion of $\frac{1+x^2}{\sqrt{1+2x}}$ in ascending powers of x up to and including the term in x^2 . State the set of values of x for which the expansion is valid. [3]
By giving x a suitable value, use the expansion to find $\sqrt{120}$ in exact form. [3]

- 3 Let $y = \ln(1 + \cos x)$.

(i) Show that $(1 + \cos x) \frac{d^2 y}{dx^2} + \cos x = \sin x \frac{dy}{dx}$. [2]

(ii) Find Maclaurin's series for y , up to and including the term in x^2 . [2]

(iii) Hence find Maclaurin's series for $\ln\left(\frac{1+x}{1+\cos x}\right)$ up to and including the term in x^2 . [2]

- 4 (a) Use de Moivre's Theorem to show that $\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^5 \left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)^7 = i$ [3]

(b) (i) Find the cube roots of $2(1 - i\sqrt{3})$. Give your answers exactly in the form $re^{i\theta}$. [3]

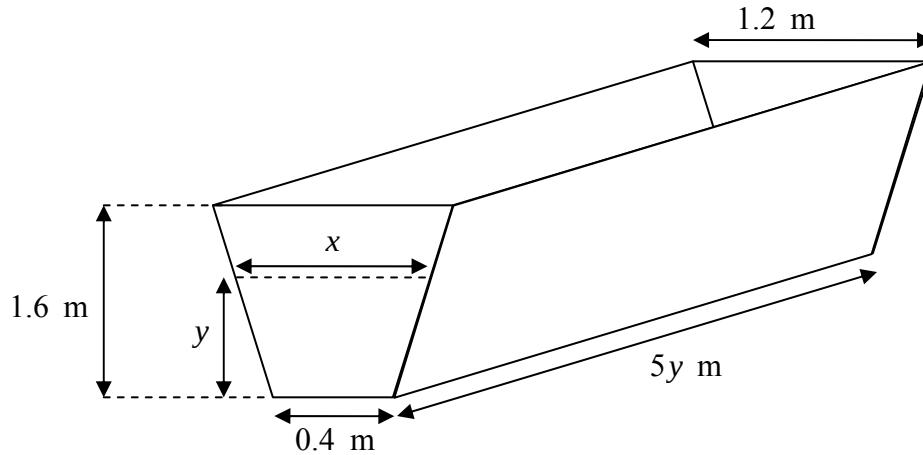
- (ii) Let the cube roots be w_1, w_2 and w_3 respectively. Plot the points W_1, W_2 and W_3 representing w_1, w_2 and w_3 in an Argand diagram. Explain why the locus of the point Z , representing the complex number z , such that $|z - w_1| = |z - w_2|$, passes through the point W_3 representing w_3 . [3]

5. Four housewives go to the supermarket together and each buy three different types of ham, picnic ham, honey-baked ham and turkey ham. The weights of the ham and the total amount paid by each of four of the housewife are shown in the following table.

	Mrs Lee	Mrs Tan	Mrs Goh	Mrs Teo
Picnic Ham (kg)	1.50	0.50	2.45	1.30
Honey-baked Ham (kg)	0.60	2.00	1.45	0.90
Turkey Ham (kg)	2.25	1.65	0.80	2.10
Total amount paid in \$	4.89	4.87	4.79	M

Assuming that, for each type of ham, the price per kilogram paid by each housewife is the same. Calculate the total amount M that Mrs Teo paid. [6]

- 6 The diagram below shows a tank with dimensions 5y m, 1.2 m, 0.4 m and 1.6 m as indicated. A liquid is poured into the tank at a rate of $0.05 \text{ m}^3/\text{s}$. At time t seconds, the liquid in the tank has a depth of y m and surface width of x m.



- (i) Using similar triangles, show that $x = \frac{y}{2} + 0.4$ and hence show that $V = \frac{y^2}{4}(5y + 8)$, where V is the volume of liquid in the tank at time t . [4]
- (ii) Initially, the tank is filled with 0.5 m^3 of liquid. Find the rate of change of the liquid depth at time $t = 55\text{s}$. [4]
- (iii) Hence find the rate of change of the surface width of the liquid at time $t = 55\text{s}$. [2]
- 7 The line l has equation $\mathbf{r} = (3\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}) + t(2\mathbf{i} + 2\mathbf{j} + \mathbf{k})$ and the point P has position vector $9\mathbf{i} + 4\mathbf{j}$.

- (i) Find an equation of the plane Π_1 which contains both l and P , giving your answer in the form $\mathbf{r} \cdot \mathbf{n} = d$. [3]

The plane Π_2 has equation $\mathbf{r} \cdot (2\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = 3$. Find

- (ii) the acute angle between Π_1 and Π_2 , correct to the nearest degree, [2]
- (iii) the equation of the line of intersection of Π_1 and Π_2 , in the form $\mathbf{r} = \mathbf{a} + t\mathbf{b}$. [3]

- 8 (a) (i) Given that a , l and S_n are the first term, n th term and the sum of the first n terms of an arithmetic progression respectively. Express n in terms of a , l and S_n . [1]

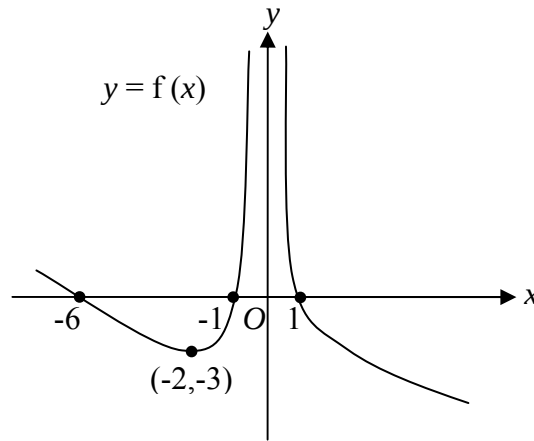
A roll of adhesive tape is wound round a circular cylinder of diameter 83 mm. The external diameter of the complete roll is 110 mm. The total length of the tape is 65800 mm.

- (ii) Let x be the thickness of the tape. By considering the length of the tape in the n^{th} layer, show that $2(n-1)x = 27$. [2]

- (iii) Find x , giving your answer correct to three significant figures. [3]

- (b) The first term of a geometric progression is 8. The sum of its first ten terms is one-eighth of the sum of the reciprocals of these terms. Show that the sum of the first seven terms of the original geometric progression is the same as the sum of the reciprocals of the seven terms. [5]

- 9 The diagram shows the graph of $y = f(x)$, which cuts the x -axis at $x = -6$, -1 and 1 and has a minimum point at $(-2, -3)$. The y -axis is the vertical asymptote.



- (a) State the range of values of x for which $f'(x) > 0$. Hence sketch the graph of $y = f'(x)$, where $f'(x)$ is the first derivative function of $f(x)$ with respect to x . [3]

- (b) Sketch, on separate clearly labeled diagrams, the graphs of
 (i) $y = f(|x|)$ (ii) $y = f(2x + 1)$ (iii) $y^2 = f(x)$ [8]

[Show on your diagrams, the equations of the asymptote, the coordinates of turning points and points of intersection with the coordinate axes.]

- 10 (a) (i)** Use the method of mathematical induction to prove that the sum of the first n terms of the series

$$\ln\left(1 - \frac{1}{4}\right) + \ln\left(1 - \frac{1}{9}\right) + \ln\left(1 - \frac{1}{16}\right) + \dots$$

$$\text{is } \ln\left(\frac{n+2}{2n+2}\right). \quad [5]$$

- (ii)** Hence, evaluate the exact value of the series. [2]

- (b) (i)** Show that $\frac{2}{r} - \frac{3}{r+1} + \frac{1}{r+2} \equiv \frac{r+4}{r(r+1)(r+2)}$. [2]

- (ii)** Hence find $S_n = \sum_{r=1}^n \frac{3r+12}{r(r+1)(r+2)}$, giving your answer in the form $k - f(n)$, where k is a constant. [4]

- (iii)** Deduce that $S_n < \frac{9}{2}$, and give a reason why S_∞ is convergent. [2]

- 11 (i)** Show that $\int_0^1 \frac{6x^4}{x^2+3} dx = 3\pi\sqrt{3} - 16$. [3]

- (ii)** The region R is the larger area bounded by the curves $y = x^2 \ln(x^2 + 3)$, $y = -x^2 + 3x$ and the line $x = 1$. Sketch the curves, showing clearly the points of intersection of the two curves and the region R . [2]

- (iii)** Using the result in **(i)**, find the exact area of R . [5]

- (iv)** Find the numerical value of the volume of revolution formed when R is rotated completely about the x -axis, giving your answer correct to 3 decimal places. [2]

End of Paper